Used Cars Price Prediction Low-Code Capstone Project

Methodology and Workflow Document

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March 30, 2026

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1. Executive Summary

The Cars4U used-car startup company intends to enter the used-car market in India. This market is characterized by rapid growth, price volatility, and notable information asymmetry. The principal mission is to develop a **machine-learning-driven pricing engine** that brings fairness, clarity, and scalability to every used car in Cars4U's inventory. The model must generate accurate, reliable, and scalable pricing while avoiding bias, price instability, and information discrepancies. A robust predictive model can significantly enhance Cars4U's market position by reducing negotiation disputes, increasing trust between buyers and sellers, and enabling automatic, data-driven pricing.

The exploratory analysis of the initial **7,253 used cars** revealed a rich, highly diverse dataset with multiple challenges including missing values and skewed distributions. Essential numerical variables (engine displacement, power, new prices) exhibited right-skewness due to luxury vehicles. Essential categorical variables (Fuel Type, Owner Type) aligned with Indian used-car market characteristics—diesel cars and first-owner vehicles comprised most of the dataset. There were over **6,247 missing values** for the New Price variable and notable gaps in Power, Engine, and Seats. Careful preprocessing and imputation strategies were required.

The modeling procedure involved multiple regression models: **Multiple Linear Regression**, **Ridge Regression**, **Lasso Regression**, **K-Nearest Neighbors (KNN) Regression**, **Decision Tree Regression**, and **Random Forest**

Regression. Linear models were interpretable but could not capture nonlinear interactions. Tree-based models best captured nonlinear interactions. The **Random Forest Regression** model had preeminent performance in R-squared, RMSE, MAE, and MAPE due to its ability to handle outliers, nonlinear relationships, and mixed variable types through its ensemble nature.

The chosen Random Forest Regression model serves as the **final solution design** for Cars4U's production pricing engine. The model aids strategic actions such as dynamic pricing, inventory refinement, and fraud detection. As a practical automated pricing engine, Cars4U can achieve price standardization, lower pricing ambiguity, aid both buyers and sellers, and establish a competitive advantage in the Indian used car market.

2. Business Problem Overview and Solution Approach

2.1 Business Problem Overview

Cars4U faces business challenges entrenched in the Indian used-car market: an inconsistent, unclear, and variable pricing atmosphere with buyers and sellers operating with incomplete information and notable regional disparities. The Indian used car market depends on authentic, dependable, and transparent pricing to combat bias, inconsistent costs, geographic diversity, and information asymmetries.

Cars4U must adopt a machine-learning pricing engine that can predict unbiased market values based on used-car attributes such as brand, model, year, and condition, empowering Cars4U to provide clear valuations, reduce negotiation conflicts, improve user confidence, and establish an adaptable, data-driven pricing strategy.

2.2 Analysis Explanation in Achieving Business Objective

A machine-learning pricing engine model allows Cars4U to navigate India's diversified used-car market, where brand reputation, geographic demand, past

ownership history, and vehicle condition interact in nonlinear ways. The model must be trained on these nonlinear relationships to reduce bias and inconsistency. This allows Cars4U to standardize used car prices across Indian cities, ensuring comparable price valuations despite local negotiation dynamics.

The pricing engine supports functioning and strategic goals by allowing accurate predictions, inventory planning, and margin enhancement. Cars4U can identify underestimated used cars for acquisition, identify overpriced listings, and predict depreciation patterns across brands and segments.

2.3 Solution Approach / Methodology

The project follows a data-driven, multi-faceted methodology using Python-based statistical modeling. The approach emphasizes Python-powered statistical analyses to deliver comprehensive data insights. The methodology centers on:

- Data cleaning and handling missing values
- Extracting brand and model information
- Feature engineering (**Car Age**, **Depreciation Amount**)
- Encoding categorical variables for modeling
- Descriptive analytics, correlation analysis, and exploratory visualizations
- Training and evaluating models: Multiple Linear Regression, Ridge Regression, Lasso Regression, KNN Regression, Decision Tree Regression, and Random Forest Regression
- Train-test split of **70/30** to ensure generalization
- Performance assessment using **R-squared**, **Adjusted R-squared**, **RMSE**, **MAE**, and **MAPE**

3. Problem Definition

3.1 Context

In 2018–2019, while new car sales totaled 3.6 million units, approximately **4 million second-hand cars** were purchased and sold. The used-car market in India is now much larger than the new-car market, necessitating data-driven pricing. Notable advantages for Cars4U include prominent market positioning, superior trust, quicker sales transactions, scalable operations, improved fraud detection and quality control, and a data advantage that compounds over time.

3.2 Objectives

Develop a trustworthy, data-driven machine-learning pricing model that accurately estimates the market value of any used car in Cars4U's inventory. Additional goals include:

- Clarity in the market
- Fair cost estimates for sellers
- Customer trust and confidence
- Competitive advantage in the Indian used-car market
- Platform competence and operational scalability
- Extensible automatic cost assessments

3.3 Key Questions

Market and Business Insights:

- What is the opportunity cost of not implementing a data-driven pricing model?
- What are the primary challenges in the Indian used-car market?
- How can the pricing engine build trust among buyers and sellers?
- What competitive advantage does an ML pricing engine provide?

Dataset Insights:

- Which features are the most influential in determining used-car prices?
- What depreciation patterns emerge across brands and segments?
- Can overpriced or underpriced patterns be detected from the data?
- What data quality issues exist, and how should they be addressed?

Modeling and Performance:

- Which model best balances accuracy and generalization?
- What are the acceptable error margins for pricing predictions?
- How does segmentation performance vary across vehicle categories?
- Can the model generalize to unseen data effectively?

Product and User Experience:

- How should predicted prices be displayed to users on the platform?
- How should the model integrate into Cars4U's existing platform?
- What explanations or breakdowns can increase trustworthiness?

Strategic and Operational:

- How can the pricing engine improve revenue and margins?
- What strategies ensure continuous model improvement?
- How durable is the competitive advantage from an ML-driven approach?

3.4 Problem Formulation

The problem is framed as a **supervised regression task**: predicting a car's fair market value from various features. The **target variable** is the price of a used car in Indian rupees (INR, expressed in lakhs). Input features include brand name, model name, location, fuel type, transmission type, manufacturing year, kilometers driven, mileage, engine displacement, power, and derived features such as age and depreciation value.

4. Data Exploration

4.1 Data Description

The dataset contains **7,253 unique used cars** with **14 distinct features**. The features are divided into categorical and numerical variables as described below.

Categorical Variables (6):

Variable	Description
S.No.	Serial number — index identifier
Name	Brand and model of the used car
Location	Cities in India (11 unique cities)
Fuel_Type	Petrol, Diesel, Electric, CNG, LPG
Transmission	Automatic, Manual
Owner_Type	First, Second, Third, Fourth & Above

Numerical Variables (8):

Variable	Description
Year	Manufacturing year (discrete)
Kilometers_Driven	Total kilometers driven by previous owner(s)
Mileage	Standard mileage in KMPL or KM/KG
Engine	Displacement volume in CC
Power	Maximum power in BHP
Seats	Number of seats (discrete)
New_Price	Price of a new car of same model (INR 100,000)
Price	Target variable — price of used car (INR 100,000)

4.2 Shape of the Dataset

The dataset comprises **7,253 rows** and **14 columns**.

4.3 Data Types

The dataset contains three data type categories: **float64** (6 columns), **int64** (3 columns), and **object** (5 columns). The S.No. column was dropped as it provides no predictive power.

4.4 Missing Values

Variable	Missing Values
S.No.	0
Name	0
Location	0
Year	0
Kilometers_Driven	0
Fuel_Type	0
Transmission	0
Owner_Type	0
Mileage	2
Engine	46
Power	175
Seats	53
New_Price	6,247
Price	1,234

The **New_Price** variable exhibits the most significant data gap with 6,247 missing values (86.1% of the dataset). The **Price** target variable has 1,234 missing values (17.0%), which required careful treatment to avoid introducing bias into the modeling process.

4.5 Duplicate Values

No duplicate entries were found across all columns in the dataset.

5. Exploratory Data Analysis (EDA)

5.1 Descriptive Statistics for Numerical Variables

Statistic	Year	Km Driven	Mileage	Engine	Power	Seats	New_Price	Price
Count	7,253	7,253	7,251	7,207	7,078	7,200	1,006	6,019
Mean	2013.37	58,699.06	18.14	1,616.57	112.77	5.28	22.78	9.48
Std	3.19	93,aborting	4.57	584.81	53.79	0.81	24.54	11.87
Min	1996	171	0.00	72	34.20	2	3.91	0.44
25%	2012	34,000	15.10	1,197	76.00	5	7.97	3.50
50%	2014	53,404	18.20	1,461	102.00	5	13.59	5.64
75%	2016	73,000	21.10	1,968	138.00	5	26.19	9.95
Max	2019	6,500,000	33.54	5,998	616	10	375	160

5.2 Descriptive Statistics for Categorical Variables

Variable	Count	Unique	Top	Freq
Name	7,253	2,041	Mahindra XUV500 W8	55

Variable	Count	Unique	Top	Freq
			2WD	
Location	7,253	11	Mumbai	949
Fuel_Type	7,253	5	Diesel	3,852
Transmission	7,253	2	Manual	5,204
Owner_Type	7,253	4	First	5,952

5.3 Univariate Analysis

Kilometers_Driven: Heavily right-skewed distribution with the majority of values falling below 100,000 km. An extreme outlier at 6,500,000 km is highly likely a data entry error. The median (53,404 km) is considerably lower than the mean (57,810.85 km), confirming the right skew. A log transformation was applied to reduce skewness and improve model stability.

Price: Heavily right-skewed with many outliers. Most used cars are priced between 0.44 and 5.64 INR lakhs, while the maximum reaches 160 INR lakhs. The mean is substantially higher than the median, indicating that luxury models drive up the average. A log transformation was applied for model stability.

Year: Most cars were manufactured between 2011 and 2016, producing a slightly left-skewed distribution. Outliers from the late 1990s and early 2000s are present. This variable directly supports the **Car_Age** feature engineering step.

Mileage: Roughly unimodal, concentrated in the 15–22 KMPL range. Some zero values were identified and treated as missing data.

Engine: Right-skewed with most values in the 1,000–1,600 CC range. A long tail extends to 6,000 CC, representing luxury and performance vehicles.

Power: Heavily right-skewed with concentration between 70–110 BHP. A long tail extends to 600 BHP for high-performance vehicles.

Seats: A heavily discrete distribution with 5-seat cars dominating at approximately 83.4% of all vehicles.

New_Price: Heavily right-skewed with most values below 50 INR lakhs. Limited to only 1,006 non-missing observations.

5.4 Bivariate Analysis

Key bivariate relationships with **Price_log** were examined:

- **Year vs. Price_log:** Clear positive correlation—newer cars command higher prices.
- **Kilometers_Driven_log vs. Price_log:** Clear negative correlation—more usage results in lower prices.
- **Engine vs. Price_log:** Clear positive relationship—larger engines are associated with higher prices.
- **Power vs. Price_log:** Strong positive relationship—higher power output correlates with premium pricing.
- **Mileage vs. Price_log:** Clear negative relationship—higher fuel efficiency (smaller, economy cars) is associated with lower prices.
- **New_Price vs. Price_log:** Strong positive relationship—one of the most potent predictors. Used cars with higher original new prices retain higher resale values.

5.5 Categorical vs. Price Analysis

Location: Mumbai, Delhi, Bangalore, and Chennai show higher median prices. Ahmedabad, Jaipur, and Kolkata show tighter, lower-priced distributions, reflecting regional economic disparities.

Fuel_Type: Diesel vehicles are the highest-priced with the widest spread. CNG, LPG, and Electric vehicles are compact and low-priced segments.

Transmission: Automatic transmission vehicles are consistently higher-priced than Manual transmission vehicles.

Owner_Type: A clear step-down hierarchy exists from First owner to Fourth & Above, reflecting diminishing value with successive ownership transfers.

Seats: 5-seat cars form a stable mid-tier price segment. 2-seat and 4-seat vehicles show higher medians, reflecting sporty and compact premium segments.

5.6 Correlation Analysis

Key correlations with **Price_log**:

Variable	Correlation with Price_log	Strength
New_Price	+0.78	Strong positive
Power	+0.77	Strong positive
Engine	+0.69	Strong positive
Mileage	−0.25	Negative
Kilometers_Driven_log	−0.21	Negative

Notable inter-feature correlations: Engine–Power (+0.86), Power–New_Price (+0.88), Engine–New_Price (+0.74). These high inter-correlations indicate multicollinearity that must be addressed during model building.

6. Feature Engineering

6.1 Brand

The **Brand** variable was extracted from the first word of the Name variable. Brand enhances reputation, authenticity, and market adjustment signals, and is considered a significantly potent categorical predictor. The top brands in the dataset are: Maruti (19.9%), Hyundai (18.5%), Honda (10.2%), and Toyota (7.0%).

6.2 Model

The **Model** variable was extracted from the second word of the Name variable. It was used primarily for hierarchical imputation of missing values in Seats and New_Price. Model was subsequently dropped from the final X feature matrix due to high cardinality, multicollinearity with Brand, and sparse categories that would produce unstable one-hot encoded features.

6.3 Car_Age

The **Car_Age** variable was calculated as:

> **Formula**
>
> Car_Age = Max Year (2019) − Manufacturing Year

A fixed reference year of 2019 ensures reproducibility. The median Car_Age is 5 years and the mean is 5.64 years. A clear negative linear relationship exists between Car_Age and Price_log—older cars consistently sell for lower prices.

6.4 Depreciation_Amount

The **Depreciation_Amount** variable was calculated as:

> **Formula**
>
> Depreciation_Amount = New_Price − Price

This feature represents the economic loss from new-to-used and summarizes brand strength, model quality, age, and mileage through value retention. The distribution is strongly right-skewed, and a strong negative linear relationship with Price_log was observed.

6.5 Updated Correlation Heatmap Insights

After feature engineering, the updated correlation structure revealed:

- **Car_Age** has a moderate negative correlation with Price_log (−0.50) and a moderate positive correlation with Kilometers_Driven_log (+0.50).
- **Depreciation_Amount** has a moderate positive correlation with New_Price (+0.48) and moderate negative correlations with Power (−0.45) and Engine (−0.43).

7. Data Preprocessing

7.1 Missing Value Treatment

Variable	Missing Count	Imputation Strategy
New_Price	6,246	Three-level hierarchical imputation: (1) median by Brand+Model, (2) median by Brand, (3) overall median
Seats	53	Two-step hierarchical: (1) median by Brand+Model, (2) overall median
Power	175	Median imputation
Engine	46	Median imputation
Mileage	Zero values as missing	Zero values treated as missing and filled with median
Price / Price_log	1,234	Rows dropped to avoid introducing bias into the target variable

7.2 Combining Infrequent Categories

Variable	Categories Combined	New Category	Combined Percentage
Fuel_Type	CNG, LPG, Electric	Other_Fuel_Type	~0.11%
Owner_Type	Third, Fourth & Above	Owner_3plus	~2.11%
Brand	14 rare brands (<1% each)	Rare_Brands	~3.3%
Seats	2.0, 6.0, 9.0, 10.0	Rare_Seat_Values	~0.9%

7.3 Outlier Handling

All outlier values were **retained** in the dataset. They represent real, meaningful used-car characteristics corresponding to luxury and performance segments. Removing or capping outliers would:

- Distort the natural distribution of the data
- Introduce artificial sanitization bias
- Reduce model transferability to new, unseen data
- Remove significant correlation patterns between luxury features and pricing

Alternative model versions built with capped outliers confirmed weaker performance, validating the decision to retain all outliers.

7.4 Key EDA and Preprocessing Insights

Most cars in the dataset are mid-age (manufactured 2011–2016), moderately driven, and powered by petrol or diesel engines. Strong right-skewed distributions are observed in Kilometers_Driven, Engine, Power, New_Price, and Price, driven by luxury outliers. Newer cars, larger engines, higher power, and higher new-car prices positively correlate with used-car prices. Higher mileage (fuel efficiency) and older age negatively correlate. Luxury brands, automatic transmission, diesel fuel type, and first-owner vehicles command higher prices across the market.

8. Building Models

8.1 Preparation

After full preprocessing, the dataset comprised **6,018 rows** and **19 columns**. The log-transformed price (**Price_log**) was used as the target variable for all models. All categorical variables were one-hot encoded, resulting in a **43-feature X matrix**. A **70/30 train-test split** was applied:

- **X_train:** 4,212 rows
- **X_test:** 1,806 rows

The following variables were dropped from the X matrix: Name (noise from high cardinality), Model (sparse dummies, multicollinearity with Brand), Kilometers_Driven (replaced by log-transformed version), and Price/Price_log (target variables).

8.2 Performance Metrics Description

Metric	Description	Interpretation
R-squared (R^2)	Proportion of variance in the target explained by the model	Higher is better (max = 1.0)
Adjusted R-squared	R-squared adjusted for the number of predictors; penalizes unnecessary features	Higher is better; closer to R^2 indicates efficient features
RMSE	Root Mean Squared Error — typical size of prediction errors (same units as target)	Lower is better
MAE	Mean Absolute Error — average absolute difference between actual and predicted	Lower is better; treats all errors equally
MAPE	Mean Absolute Percentage Error — average percentage difference	Lower is better; scale-independent

All predictions were exponentiated back to the original price scale for metric computation, ensuring business-interpretable error measurements.

8.3 Model 1: Multiple Linear Regression (OLS)

Definition: Multiple Linear Regression describes the relationship between the target and predictors by fitting a single linear equation that minimizes the sum of squared errors. It assumes a linear relationship between features and the target and finds optimal coefficients using linear algebra.

Building Process:

- Multicollinearity was addressed through iterative VIF analysis: Year (INF VIF), Engine (VIF ~10.62), Seats_5.0 (VIF ~9.35), and Brand_maruti (VIF ~7.89) were removed. All remaining features achieved VIF < 5.
- Statistically insignificant variables were removed ($p > 0.05$): Location_Chennai, Brand_hyundai, Location_Mumbai, Location_Kochi, Brand_volkswagen, Brand_nissan, and Seats_8.0.
- Final model: R-squared = 0.927 in summary output, with all p-values < 0.05.

Assumptions Verified:

1. **Mean of residuals ≈ 0:** Value of 4.13e-15 — satisfied.
2. **Homoscedasticity:** Goldfeld-Quandt test p-value = 0.19, F-statistic = 1.04 — satisfied.
3. **Linearity:** Residuals randomly scattered around zero — satisfied.
4. **Normality:** Q-Q plot and histogram confirm bell-shaped, normal residuals — satisfied.

Performance:

Metric	Train	Test
R-squared	0.45	0.20
Adjusted R-squared	0.45	0.19
RMSE	8.29	9.97
MAE	2.06	2.11
MAPE	0.18	0.18

The model demonstrates modest explanatory power with evidence of underfitting and poor generalization to unseen data.

8.4 Model 2: Ridge Regression (L2 Regularization)

Definition: Ridge Regression extends Multiple Linear Regression with an L2 penalty that increases with the square of weights, encouraging small but nonzero weights. It prevents overfitting and stabilizes coefficients, particularly in the presence of multicollinearity. Ridge Regression is sensitive to feature scaling.

Building Process: Two versions were created—one with the initial X matrix and one without the Year variable (INF VIF). The second version was chosen for slightly higher adjusted R-squared on the training data.

Performance:

Metric	Train	Test
R-squared	0.51	0.31
Adjusted R-squared	0.51	0.29
RMSE	7.78	9.26
MAE	1.99	2.01
MAPE	0.18	0.18

Ridge Regression shows moderate explanatory power, an improvement over MLR, and consistent error behavior across training and test datasets.

8.5 Model 3: Lasso Regression (L1 Regularization)

Definition: Lasso Regression extends Multiple Linear Regression with an L1 penalty that penalizes nonzero weights, setting some coefficients to exactly zero. It performs regularization and feature selection simultaneously, effectively identifying the most important predictors.

Building Process: Three versions were built—the initial version (negative R-squared), a hyperparameter-tuned version with alpha = 0.1, and one with alpha = 0.2. The alpha = 0.1 version was chosen as the final model.

Performance:

Metric	Train	Test
R-squared	0.52	0.65
Adjusted R-squared	0.52	0.64
RMSE	7.72	6.61
MAE	2.94	2.91
MAPE	0.29	0.28

Lasso Regression demonstrates solid explanatory power. The test R-squared being higher than the training R-squared indicates that L1 regularization is effectively filtering noise. Stable predictive behavior is observed across both datasets.

8.6 Model 4: Decision Tree Regression

Definition: Decision Tree Regression recursively splits data into regions based on feature thresholds. It captures nonlinear patterns and interactions without requiring feature transformations. Feature importance is measured by **Gini Importance** (variance reduction across splits).

Building Process: Four versions were created with varying max_depth and min_samples_leaf parameters. **RandomizedSearchCV** was used for hyperparameter tuning. The final model used max_depth = 5 and min_samples_leaf = 5.

Top Features (Gini Importance): Power, Depreciation Amount, New Price, Year, Engine.

Performance:

Metric	Train	Test
R-squared	0.96	0.86
Adjusted R-squared	0.96	0.86
RMSE	2.29	4.20
MAE	1.05	1.21
MAPE	0.12	0.13

The Decision Tree model delivers strong performance with some overfitting (10-point gap in R-squared between train and test). Low MAE and MAPE values are maintained across both datasets.

8.7 Model 5: Random Forest Regression

Definition: Random Forest Regression builds on the Decision Tree process by averaging predictions from multiple randomized Decision Trees. Each tree is trained on bootstrapped samples with random feature subsets. This ensemble approach reduces overfitting and improves stability through bootstrap aggregation (bagging).

Building Process: Four versions were built with varying parameters. **RandomizedSearchCV** was used for hyperparameter tuning. The third version was rejected due to an R-squared of 1.0 on training data (severe overfitting). The fourth version was selected as the final model with max_depth = 5.

Top Features (Gini Importance): Power, Depreciation Amount, New Price, Year, Car Age, Kilometers_Driven_log.

Performance:

Metric	Train	Test
R-squared	**0.97**	**0.90**
Adjusted R-squared	**0.97**	**0.90**
RMSE	**2.07**	**3.59**
MAE	**0.79**	**0.88**
MAPE	**0.09**	**0.09**

Random Forest Regression achieves the **strongest performance across all metrics**. The model maintains a healthy balance between accuracy and generalization with only a 7-point gap in R-squared. Identical MAPE on training and test datasets (0.09) indicates **exceptional stability** and robust generalization.

8.8 Model 6: KNN Regression

Definition: K-Nearest Neighbors Regression is a non-parametric, instance-based method that predicts by averaging the target values of the k most similar

observations. It is flexible but sensitive to feature scaling and local density variations.

Building Process: Two versions were evaluated—the default (5 neighbors) and a 10-neighbor variant. The default version was chosen for slightly higher test R-squared. Features were scaled before training.

Performance:

Metric	Train	Test
R-squared	0.89	0.80
Adjusted R-squared	0.89	0.80
RMSE	3.66	5.04
MAE	1.79	2.29
MAPE	0.20	0.25

KNN Regression demonstrates robust yet moderate performance. The model is sensitive to noise and shows a healthy balance between generalization and error behavior, though it does not match tree-based models in overall accuracy.

9. Models and Techniques Comparison

9.1 Key Measures of Success

All models are evaluated on five key performance metrics: **R-squared**, **Adjusted R-squared**, **RMSE**, **MAE**, and **MAPE**. Test dataset metrics are considered more important than training dataset metrics, as they reflect the model's ability to generalize to unseen data.

9.2 Comprehensive Performance Comparison Table

Model	Train R^2	Test R^2	Train Adj R^2	Test Adj R^2	Train RMSE	Test RMSE	Train MAE	Test MAE	Train MAPE	Test MAPE
Linear Regression	0.45	0.20	0.45	0.19	8.29	9.97	2.06	2.11	0.18	0.18
Ridge Regression	0.51	0.31	0.50	0.29	7.78	9.25	1.99	2.01	0.18	0.18
Lasso Regression	0.52	0.65	0.52	0.64	7.72	6.61	2.94	2.91	0.29	0.28
Decision Tree	0.96	0.86	0.96	0.86	2.29	4.20	1.05	1.21	0.12	0.13
Random Forest	**0.97**	**0.90**	**0.97**	**0.90**	**2.07**	**3.59**	**0.79**	**0.88**	**0.09**	**0.09**
KNN Regression	0.89	0.80	0.89	0.80	3.66	5.04	1.79	2.29	0.20	0.25

9.3 Performance Analysis

Linear models (Multiple Linear Regression, Ridge Regression, Lasso Regression) are fundamentally limited by their linear assumptions and cannot capture the nonlinear relationships inherent in used-car pricing. Their R-squared values on the test set range from 0.20 to 0.65.

Tree-based models significantly outperform linear models by capturing complex, nonlinear interactions between features without requiring explicit transformations.

Random Forest Regression emerges as the best model across all metrics:

- Highest test R-squared: **0.90**
- Lowest test RMSE: **3.59**
- Lowest test MAE: **0.88**
- Lowest test MAPE: **0.09**

The reasons for Random Forest superiority include:

5. Captures nonlinear relationships between vehicle attributes and price
6. Reduces overfitting through bootstrap aggregation (bagging)
7. Handles high-dimensional mixed data (numerical and categorical) efficiently
8. Generates durable, low-error predictions across diverse vehicle segments
9. Avoids the instability of a single Decision Tree through ensemble averaging

10. Final Solution Design

10.1 Model Adopted

Random Forest Regression is selected as the final production model for Cars4U's pricing engine. The model was trained on the fully preprocessed dataset of **6,018 used cars**, utilizing a **43-feature X_train matrix** and a **log-transformed Price target**.

10.2 Expected Benefits

- R-squared of **0.90** on the test dataset with the lowest error metrics across all evaluated models
- Optimal robustness to nonlinear relationships and complex feature interactions
- Superior bias-variance tradeoff compared to a single Decision Tree

- Meets practical business fit criteria for generating accurate, reliable used-car prices

10.3 Costs and Tradeoffs

- **Higher computational cost:** Increased training time, memory requirements, and hyperparameter tuning complexity
- **Lower interpretability:** Less transparent than linear models; ensemble nature makes individual predictions harder to explain
- **Deployment complexity:** More complex to deploy, monitor, and maintain in production environments
- **Feature sensitivity:** Can be sensitive to irrelevant or noisy features in the input matrix
- **Larger model size:** Requires more storage than simpler regression models

10.4 Production Readiness and Interpretability

The model's high R-squared (0.90), low RMSE, low MAE, and stable MAPE make it viable for production deployment. Global feature importance analysis identifies the key price drivers:

10. **Power** — Maximum engine output in BHP
11. **Depreciation Amount** — Value loss from new to used
12. **New Price** — Original new-car price
13. **Year** — Manufacturing year
14. **Car Age** — Years since manufacture
15. **Kilometers Driven (log)** — Log-transformed usage metric

The Random Forest model is the quickest for real-time pricing, the most robust to moderate data drift, and the most adaptable to standardized versioning, monitoring, and retraining workflows.

11. Conclusions and Recommendations

11.1 Key Findings

- India's used car market offers high growth potential but poses significant configurational pricing challenges due to regional disparities and information asymmetry.
- The dataset depicts unreliable valuations, regional discrepancies, and significant information gaps — particularly in the New_Price variable (86.1% missing).
- Exploratory data analysis reveals strong predictive signals from several variables (Power, New_Price, Engine) but considerable data quality issues that require rigorous preprocessing.
- Nonlinear machine learning models substantially outperform linear models for used-car pricing tasks.
- Random Forest Regression achieves the best performance across all evaluation metrics: highest R-squared, lowest RMSE, lowest MAE, and lowest MAPE.
- The top predictors identified are: Power, Depreciation Amount, New Price, Year, Car Age, and Kilometers Driven (log).

11.2 Key Takeaways

- A machine-learning pricing engine is a **critical business requirement** for Cars4U to compete effectively in the Indian used-car market.
- Feature engineering (Car Age, Depreciation Amount, Brand extraction) **substantially enhances** predictive robustness and model interpretability.
- The Random Forest Regression model supports **dynamic pricing**, **inventory optimization**, and **fraud detection** as integrated strategic capabilities.

- Continuous data enrichment and regular model retraining are essential for sustained competitive advantage.

11.3 Business Recommendations

1. Improve Data Quality: Standardize data-collection procedures across all intake channels. Mandate required fields for critical variables (Engine, Power, New_Price). Implement validation rules at the point of entry and deploy automated irregularity detection to flag suspicious submissions.

2. Operational Integration: Integrate model outputs into seller listings, listing formation workflows, and buyer-facing price displays. Use predicted prices to inform procurement decisions, identify undervalued acquisition targets, and flag overpriced listings for review.

3. Periodic Model Retraining: Adopt a systematic retraining schedule with continuous data enrichment. Monitor model drift through tracking error metrics over time. Maintain a feedback loop that incorporates actual transaction prices into subsequent training cycles.

4. Strategic Applications: Leverage depreciation patterns and brand-specific value retention insights for inventory mix optimization. Use regional price trends for targeted marketing campaigns. Support fraud detection by flagging listings with anomalous pricing relative to model predictions.

5. Transparent Communication: Provide buyers and sellers with streamlined explanations of key pricing factors. Offer visual breakdowns of how individual features contribute to the predicted price. Display confidence ranges alongside price estimates to set appropriate expectations.

11.4 Conclusions

Cars4U can effectively transform its pricing operations through the Random Forest Regression model. The combination of strong predictive performance ($R^2 = 0.90$, MAPE = 0.09), explainability through feature importance, and scalability through ensemble architecture makes it best suited for real-world deployment. Long-term success depends on sustaining data quality, incorporating model insights into business techniques, and treating pricing intelligence as a strategic pillar that underpins competitive differentiation in the Indian used-car market.

12. Workflow Summary

The following outlines the end-to-end project pipeline from problem formulation through final delivery.

Phase 1: Problem Formulation

- Identified the task as a **supervised regression problem**
- Defined the target variable: Price (INR 100,000 / lakhs)
- Established key business questions and objectives for Cars4U's market entry

Phase 2: Data Collection and Exploration

- Loaded dataset: **7,253 rows**, **14 columns**
- Identified 6 categorical and 8 numerical variables
- Assessed missing values (most critically, 6,247 missing in New_Price), duplicates, and data types
- Identified extreme values (e.g., 6.5 million km outlier, zero-mileage entries)

Phase 3: Exploratory Data Analysis

- Conducted **univariate analysis** (histograms, boxplots, bar plots) for all variables
- Performed **bivariate analysis** (scatterplots with regression lines) for numerical variables vs. Price_log
- Analyzed **categorical vs. Price** relationships (boxplots with and without outliers)
- Generated **correlation heatmap** analysis to identify multicollinearity and predictive strength

Phase 4: Feature Engineering

- Extracted **Brand** and **Model** from the Name variable
- Created **Car_Age** from the Year variable (reference year: 2019)
- Created **Depreciation_Amount** from New_Price minus Price
- Updated correlation analysis to incorporate new engineered features

Phase 5: Data Preprocessing

- Applied hierarchical imputation for New_Price (Brand+Model → Brand → overall median) and Seats
- Applied median imputation for Engine, Power, and Mileage
- Dropped 1,234 rows with missing Price values to protect target variable integrity
- Combined infrequent categories in Fuel_Type, Owner_Type, Brand, and Seats
- Retained all outliers as meaningful market signals
- One-hot encoded categorical variables (resulting in 43 features)
- Applied **70/30 train-test split** (4,212 train / 1,806 test)

Phase 6: Model Building and Evaluation

- Built and evaluated **6 regression models**: MLR, Ridge, Lasso, Decision Tree, Random Forest, KNN
- Applied multicollinearity diagnostics (VIF) for linear models
- Performed hyperparameter tuning via **RandomizedSearchCV** for tree-based models
- Verified linear regression assumptions (residual mean, homoscedasticity, linearity, normality)
- Compared all models on R-squared, Adjusted R-squared, RMSE, MAE, and MAPE

Phase 7: Final Solution Design

- Selected **Random Forest Regression** as the production model (Test R^2 = 0.90, MAPE = 0.09)
- Documented expected benefits, computational costs, and interpretability tradeoffs
- Assessed production readiness and real-time pricing viability

Phase 8: Business Recommendations and Conclusions

- Formulated recommendations for data quality, operational integration, retraining, and strategic applications
- Established pricing intelligence as a **strategic business pillar** for Cars4U's competitive differentiation

Summary

This document presents the complete methodology and workflow for the Cars4U Used Cars Price Prediction Capstone Project. The Random Forest Regression model, with an R-squared of 0.90 and MAPE of 0.09, is recommended as the production pricing engine. The model enables accurate, scalable, and transparent used-car pricing to support Cars4U's entry into the Indian market.

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